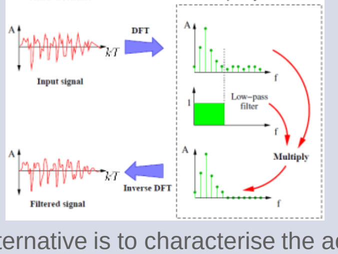
COM3502/4502/6502 SPEECH PROCESSING

Lecture 7

The 'Z' Transform

Filtering (Frequency Domain)

- A signal may be filtered by
  1. transforming it into the frequency domain (*using DFT*)
  2. multiplying the signal spectrum by the filter spectrum $y[n] = x[n]h[n]$
  3. transforming it back into the time domain (*using IDFT*)



- An alternative is to characterise the action of a filter in the time domain (*i.e. on the waveform itself*)
- This is done using 'difference equations'

Filtering (Time Domain)

- Consider a linear filter where …
  - $x[kT] := x[k]$ is the input
  - $y[kT] := y[k]$ is the output (*response*)

Omitting sampling interval T for simplicity; *k*: discrete time index

Input Signal $x[k] \longrightarrow \boxed{h[k]} \longrightarrow y[k]$ Output Signal

- 'Difference equations' relate the current filter output to the current and past inputs *and* outputs
- Design of filters can be done in the 'Z-Domain'

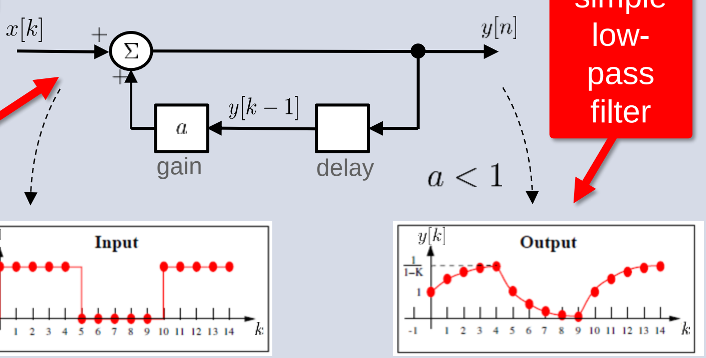
Difference Equations: *an example*

Input Signal $x[k] \longrightarrow \boxed{\text{Filter } h[k]} \longrightarrow y[k]$ Output Signal

Difference equation

$$y[k] = a\,y[k-1] + x[k]$$

Equivalent processing diagram



Result is a simple low-pass filter

Linear Time-Invariant Filters

- A '**linear filter**' obeys the principle of super-position …
  - if $x_1[k]$ yields $y_1[k]$
  - and input is $a\ x_1[k] + b\ x_2[k]$
  - then $y[n] = a\ y_1[k] + b\ y_2[k]$
- The response of a '**time-invariant filter**' is the same for all times …
  - if $x[k]$ yields $y[k]$
  - then $x[k\text{-}k_0]$ yields $y[k\text{-}k_0]$

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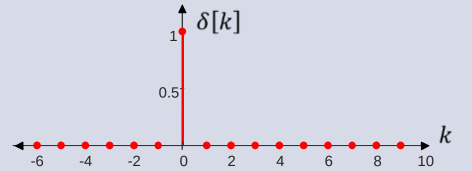
Linear Time-Invariant Filters

- Linearity and time-invariance allows the input (*and the response*) of a filter to be constructed from a weighted sum of time-shifted impulses $\delta\,[k]$

  where $\delta\,[k]$ is 1 when $n = 0$ (and 0 otherwise)



- Time-invariance and super-position thus allows the input signal to be expressed as …

$$x[k] = \sum_{i=-\infty}^{\infty} x[i]\,.\,\delta[k-i]$$

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Linear Time-Invariant Filters

- Given the '**impulse response**' of a filter $h[k]$, the output is …

$$y[k] = \sum_{i=-\infty}^{\infty} x[k-i]\ h[i]$$

- This is the '**discrete convolution**' of the input signal with the filter impulse response, and it can be written as …

$$y[k] = x[k] * h[k]$$

convolution operator

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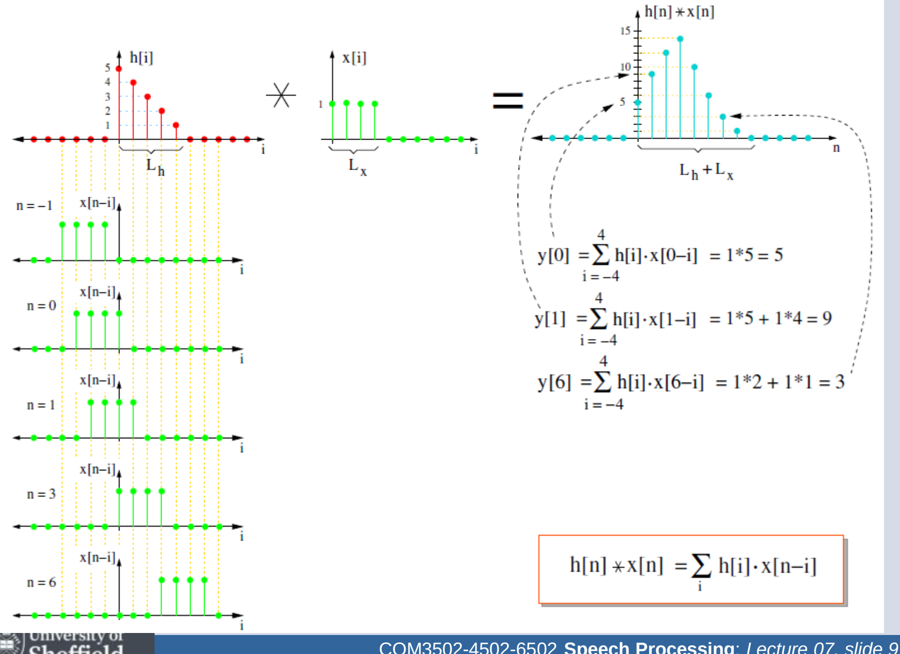

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The Z-Transform

- Analysis of filters in the time domain (*i.e. using difference equations*) is cumbersome
- The 'Z-transform' is a power series representation of a discrete-time sequence
- E.g. for the sequence $x[0]\ x[1]\ x[2]\ x[3]$, the Z transform simply multiplies each coefficient in the sequence by a power of z corresponding to its index …

$$X(z) = x[0]\,z^{0} + x[1]\,z^{-1} + x[2]\,z^{-2} + x[3]\,z^{-3}$$

- By convention, *negative* powers of z are used for *positive* time indices (*i.e.* z^{-k} *represents a delay of* k *samples*)

The Z-Transform

- For a general sequence, the Z transform is written as …

$$X(z) = \sum_{n=-\infty}^{\infty} x[k]\, z^{-k}$$

- The transform is denoted by the $Z\{\ \dots\ \}$ operator …

$$X(z) = Z\{x[k]\}$$

The Z-Transform

Key properties of the transform are …

– linearity

$$Z\{ax_1[k] + bx_2[k]\} = a\,Z\{x_1[k]\} + b\,Z\{x_2[k]\}$$

– time-shifting

$$Z\{x[k-k_0]\} = z^{-k_0} Z\{x[k]\}$$

– convolution

$$Z\{x[k] * y[k]\} = Z\{x[k]\}\,Z\{y[k]\}$$

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The Z-Transform

- As we saw, the output of a linear filter is described by the discrete convolution …

$$y[k] = x[k] * h[k]$$

- This can be expressed using the Z transform as a product …

$$Y[z] = X[z]\,H[z]$$

- Hence the 'z-plane transfer function' $H[z]$ may be written as …

$$H[z] = \frac{Y[z]}{X[z]}$$

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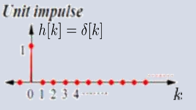
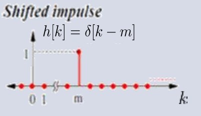
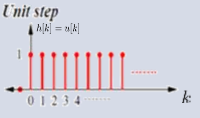
Common Z-Transform Pairs



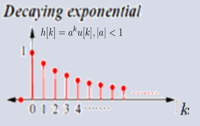
$\Rightarrow$ $$H[z]=1$$



$\Rightarrow$ $$H[z]=z^{-m}$$



$\Rightarrow$ $$H[z]=\frac{1}{1-z^{-1}}$$



$\Rightarrow$ $$H[z]=\frac{1}{1-a\cdot z^{-1}}$$

Link with the Fourier Transform

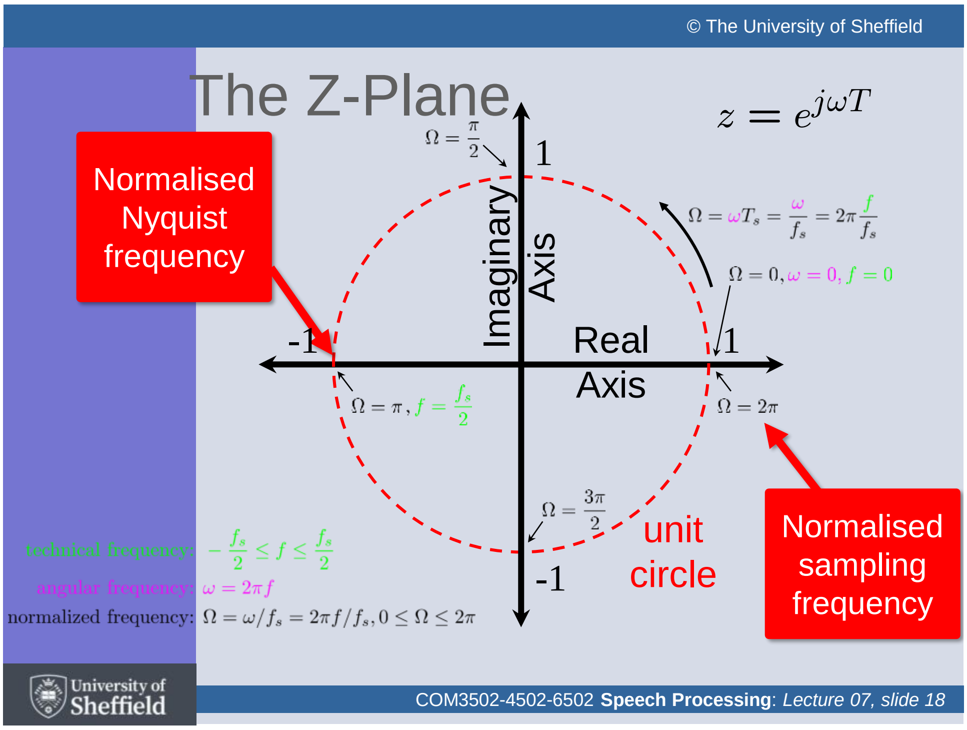
Z-Transform

$$Z\{x[k]\}=\sum_{k=-\infty}^{\infty}x[k]z^{-k}$$

- If $z>1$, then the transform is equivalent to multiplying the sequence by a *rising* exponential
- If $0<z<1$, then the transform is equivalent to multiplying the sequence by a *falling* exponential
- If z is a complex number *lying on the unit circle*, then the transform is equivalent to multiplying the sequence by a real cosine and imaginary sinusoid

  i.e. this is directly equivalent to the Fourier transform!
- Hence, given a filter transfer function $H[z]$, the filter frequency response can be evaluated at any frequency ω by setting …

$$z=e^{j\omega T}$$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The Z-Plane
$z = e^{j\omega T}$
Normalised Nyquist frequency
$\Omega = \frac{\pi}{2}$
1
Imaginary Axis
$\Omega = \omega T_s = \frac{\omega}{f_s} = 2\pi\frac{f}{f_s}$
$\Omega = 0, \omega = 0, f = 0$
-1
Real Axis
1
$\Omega = \pi, f = \frac{f_s}{2}$
$\Omega = 2\pi$
$\Omega = \frac{3\pi}{2}$
unit circle
-1
Normalised sampling frequency
technical frequency: $-\frac{f_s}{2} \leq f \leq \frac{f_s}{2}$
angular frequency: $\omega = 2\pi f$
normalized frequency: $\Omega = \omega/f_s = 2\pi f/f_s, 0 \leq \Omega \leq 2\pi$
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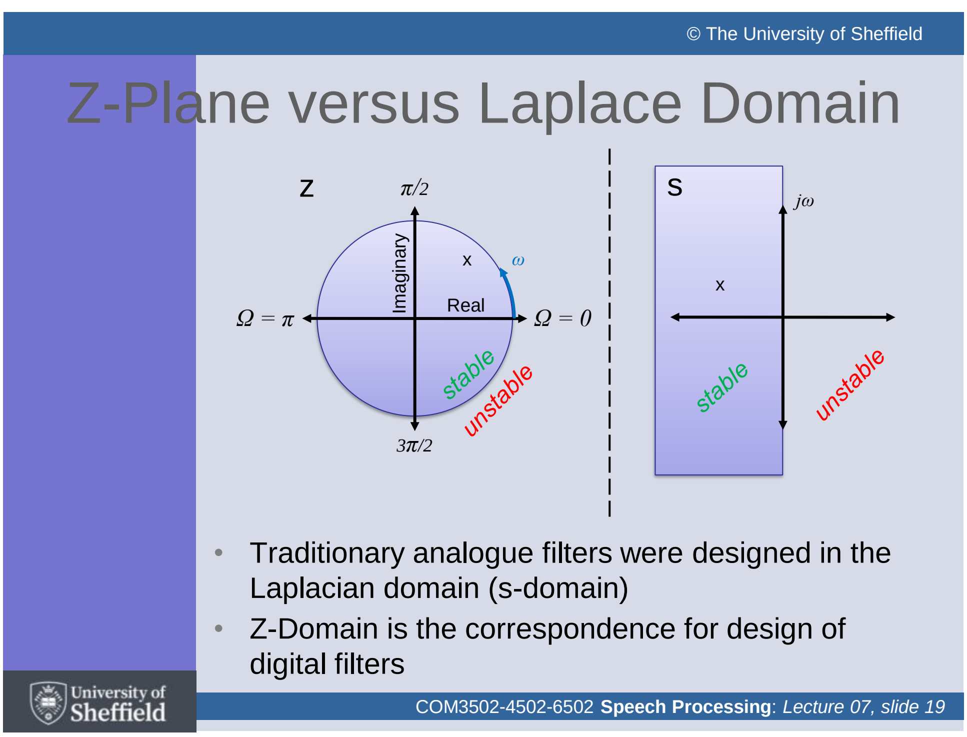
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Z-Plane versus Laplace Domain
Z
$\pi/2$
Imaginary
x
ω
Real
$\Omega = \pi$
$\Omega = 0$
stable
unstable
$3\pi/2$
S
$j\omega$
x
stable
unstable
• Traditionary analogue filters were designed in the Laplacian domain (s-domain)
• Z-Domain is the correspondence for design of digital filters
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Poles and Zeros

- Linear filter transfer functions (*arising from general difference equations*) can be written as the ratio of two polynomials in z …

$$H[z]=\frac{P[z]}{Q[z]}$$

- The values of z for which $P[z]=0$ are known as the '**zeros**' of $H[z]$
- The values of z for which $Q[z]=0$ are known as the '**poles**' of $H[z]$
- Poles correspond to frequencies at which a filter transfer function tends to infinity (*i.e. 'resonances'*)
- Zeros correspond to frequencies at which a filter transfer function tends to zero (*i.e. 'anti-resonances'*)
- A filter is completely characterised by its poles and zeros

Poles and Zeros

- The z-domain polynomials are often written in terms of their poles and zeros z_i …

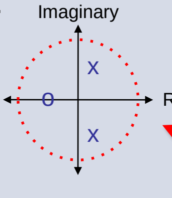
Zeros

$$P[z]=\prod_{i=1}^{p}(z-z_i) \qquad Q[z]=\prod_{i=1}^{q}(z-z_i)$$

Poles

- The poles and zeros correspond to the '**roots**' of the difference equations
- They can be visualised by plotting their positions in the '**Z-Plane**' …

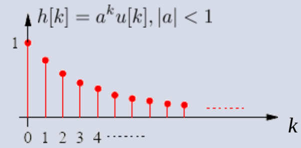
Zeros are signified by 'o's



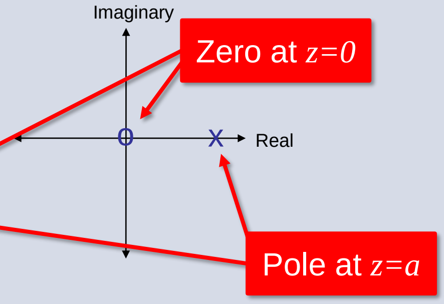
Poles are signified by 'x's

Poles and Zeros: *an example*

$$h[k] = 0 \quad \text{for } k < 0$$
$$h[k] = a^k \quad \text{otherwise}$$



$$H[z] = \sum_{n=-\infty}^{\infty} h[k]\, z^{-k} = \sum_{k=0}^{\infty} a^k\, z^{-k} = \sum_{k=0}^{\infty} (a\, z^{-1})^k = \frac{1}{1 - a\, z^{-1}} = \frac{z}{z - a}$$



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Poles and Zeros

- The magnitude response of a filter can be quickly understood based on the location of its poles and zeros
- By starting with a pole/zero plot, it is possible to <u>design</u> a filter and obtain its transfer function very easily …

$$H[z] = \frac{\prod_{i=1}^{p} (z - z_i)}{\prod_{i=1}^{q} (z - z_i)} = \frac{\prod \text{"distance of zeros from unit circle"}}{\prod \text{"distance of poles from unit circle"}}$$

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Filter Frequency Response

- The frequency response of a filter can be plotted using the distances from the 'unit circle' to the poles and zeros
- While moving around the unit circle …
  - if close to a zero, then the magnitude is small
  - if close to a pole, then the magnitude is large
- If a *zero* is on the unit circle, then the frequency response is zero at that point
- If a *pole* is on the unit circle, then the frequency response goes to infinity at that point

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Filter Response: *an example*

- Consider the simple averaging filter with the following difference equation …

$$y[k] = ay[k-1] + x[k]$$

- Taking Z transforms, this becomes …

$$Y(z) = aY(z)z^{-1} + X(z)$$

- Hence the transfer function ($Y(z)/X(z)$) is given by …

$$Y(z) - aY(z)z^{-1} = X(z)$$
$$Y(z)(1 - az^{-1}) = X(z)$$
$$\frac{Y(z)}{X(z)} = \frac{1}{1 - az^{-1}}$$

Zero at z=0

Pole at z=a

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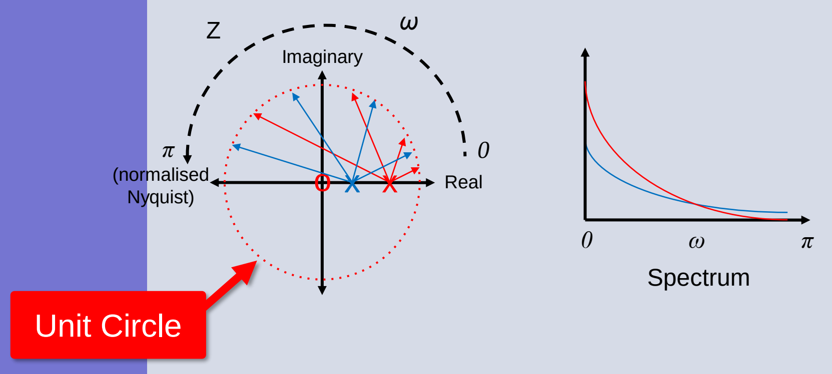
Filter Response: *real pole*



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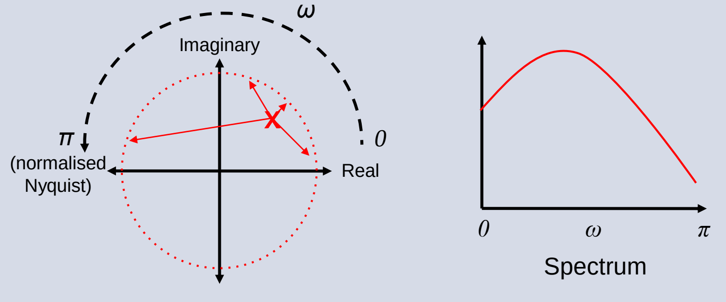
Filter Response: *complex pole*



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This slide set has covered …

- Filtering
- Difference equations
- Convolution
- The Z transform
- Link with the Fourier transform
- Poles and zeros
- The Z Plane

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Any Questions ?

Raise during lecture or post in the Blackboard Discussion group

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Next time …

The Fourier Transform